Kernix 0.10.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

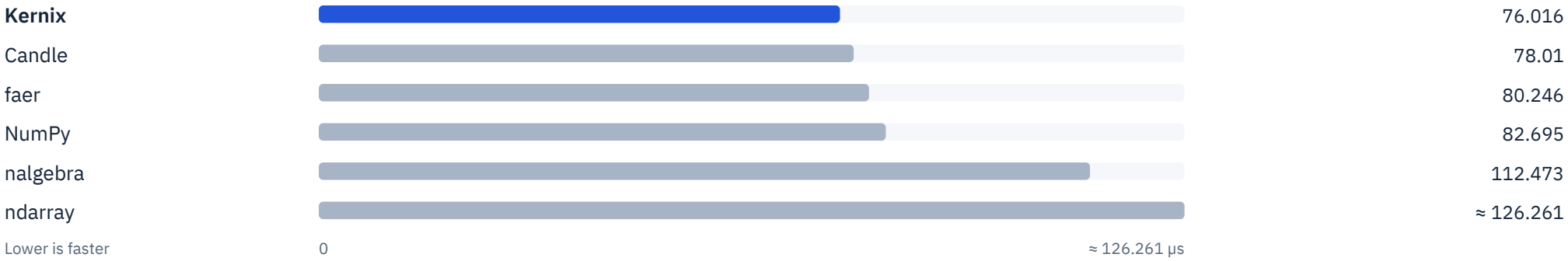
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



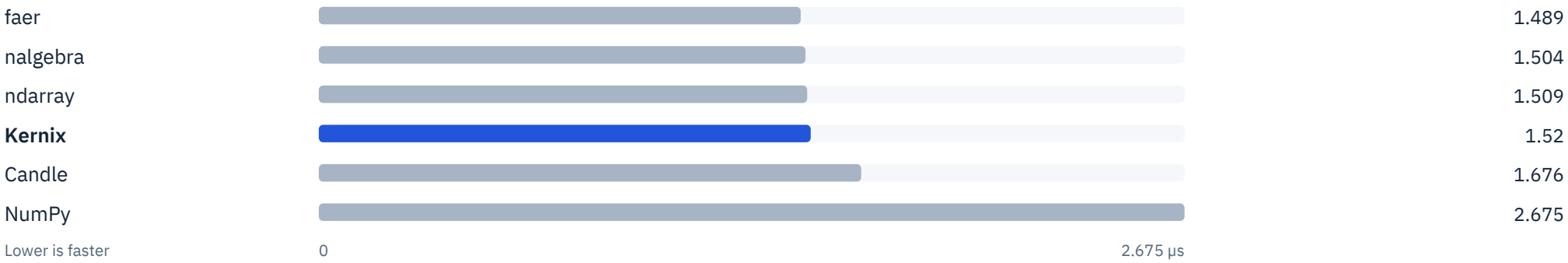
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

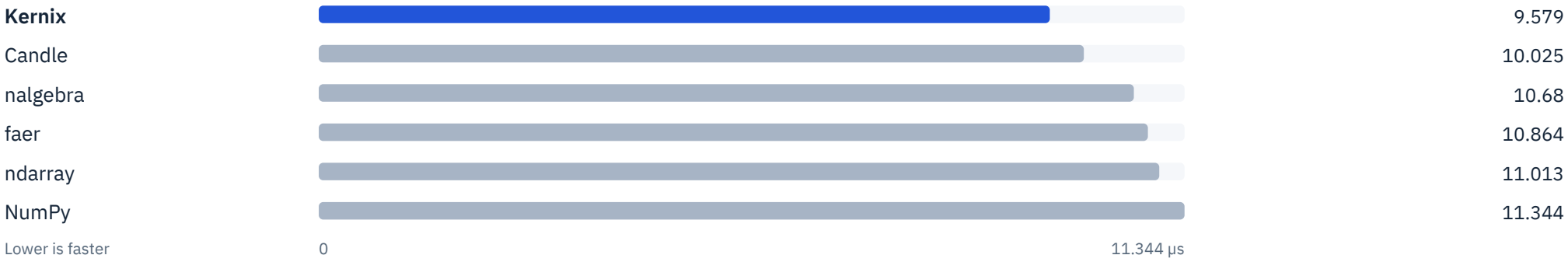
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



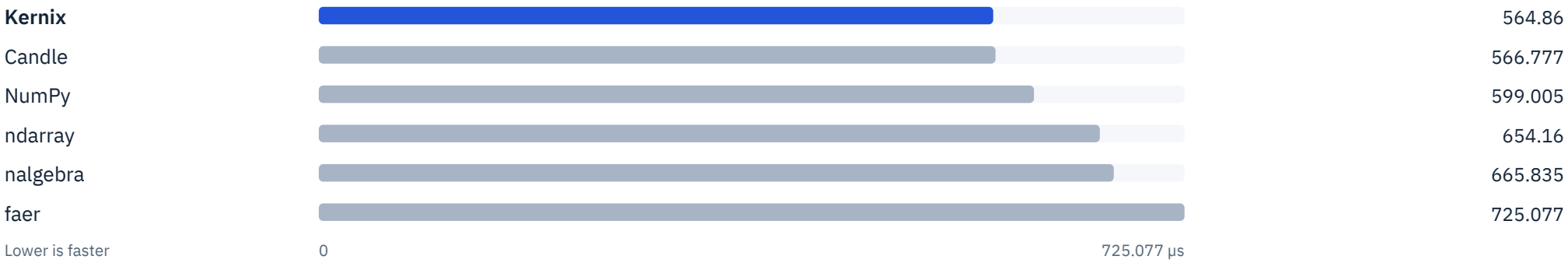
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

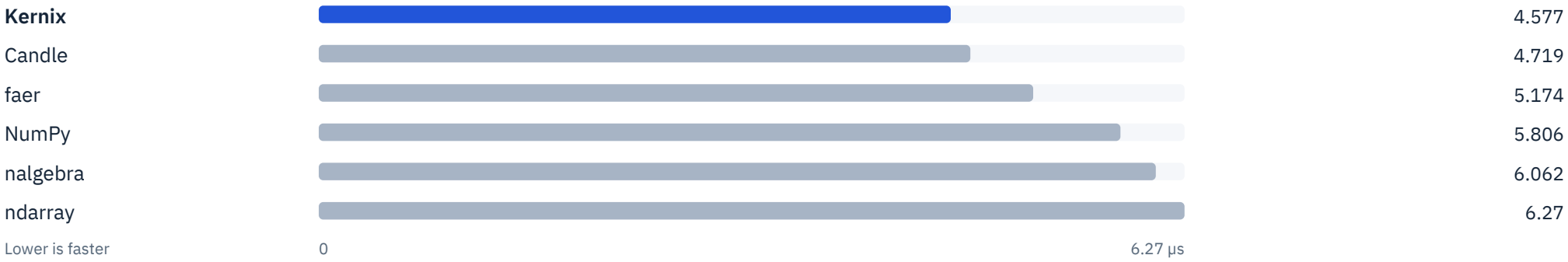
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

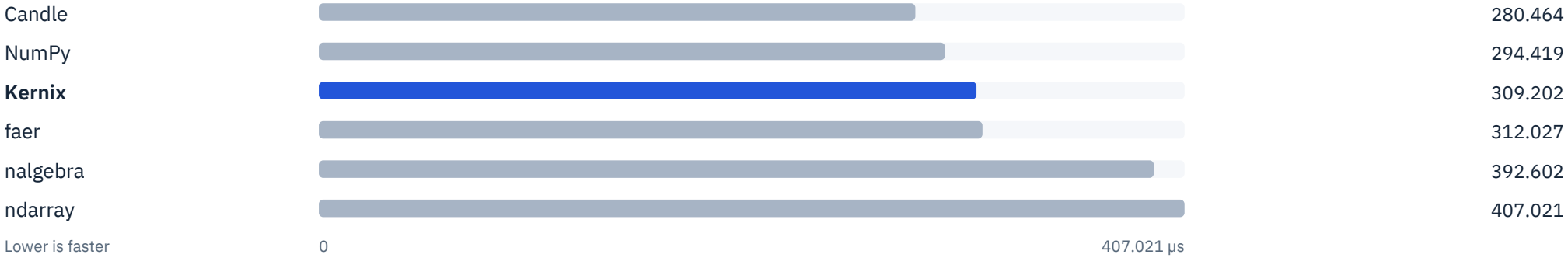
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



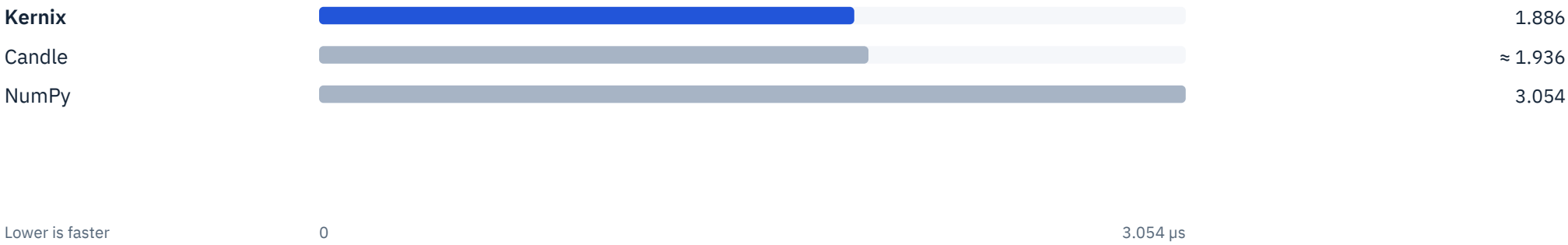
Kernix 0.10.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

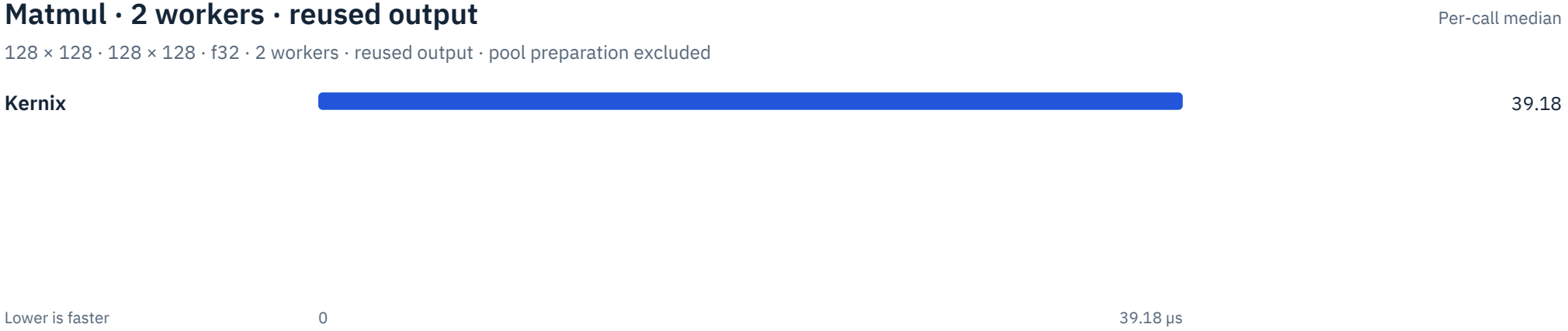
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



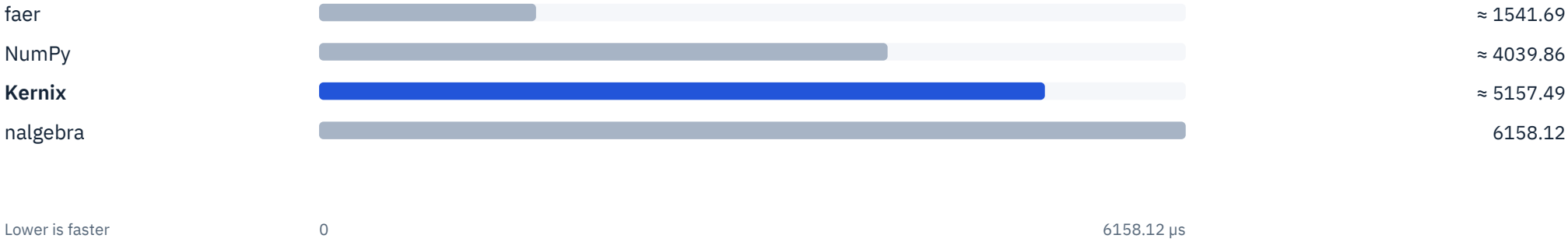
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



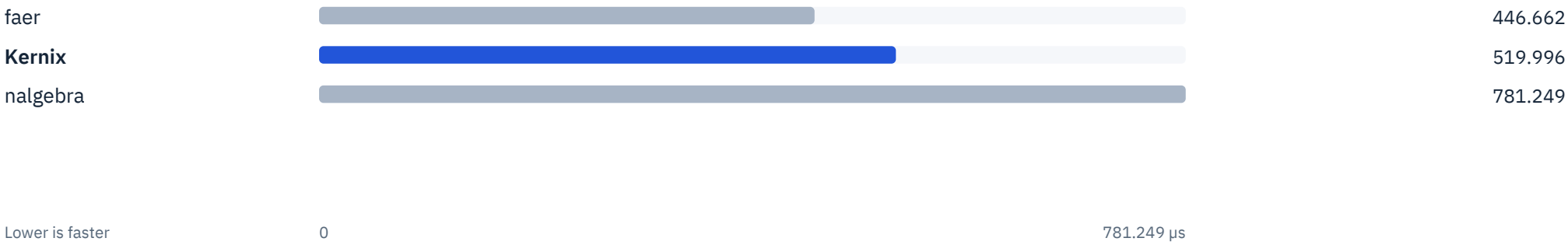
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

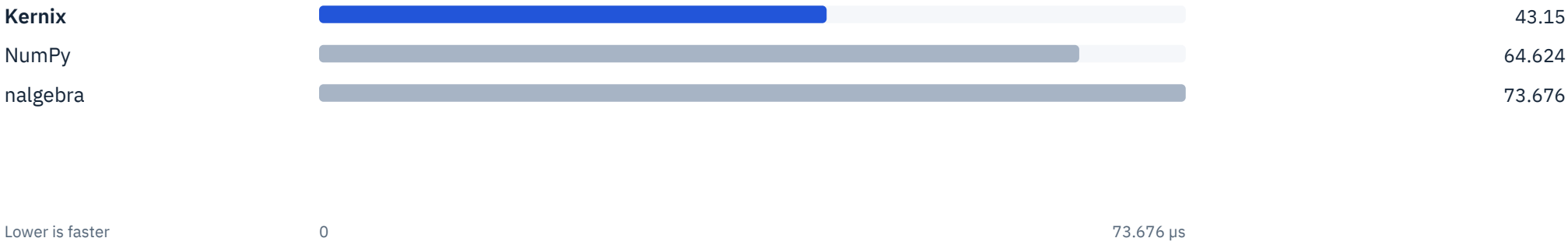
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



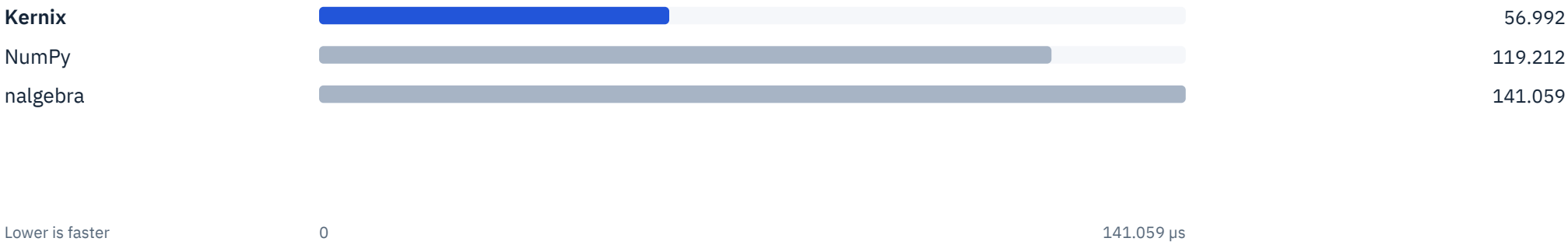
Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

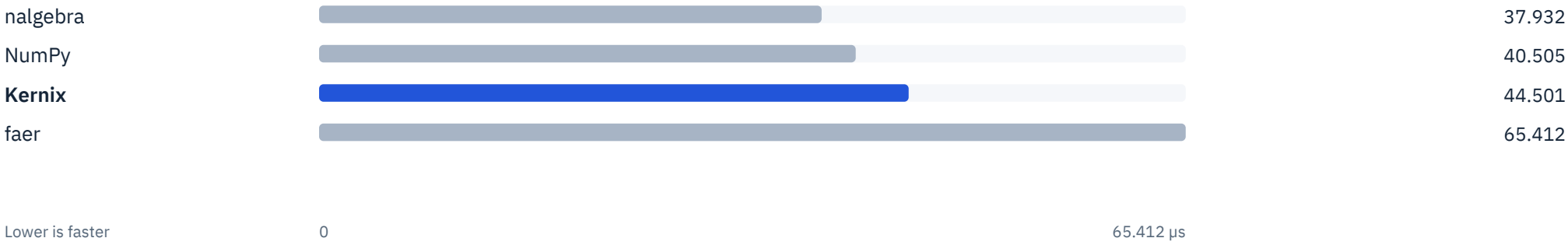
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

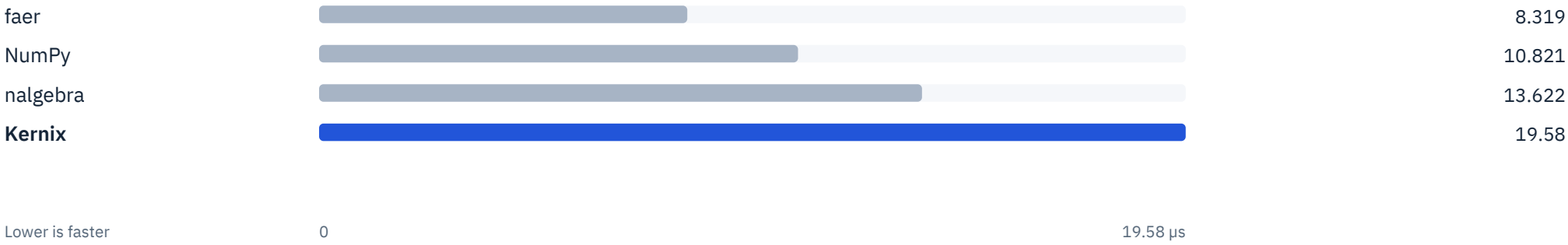
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median

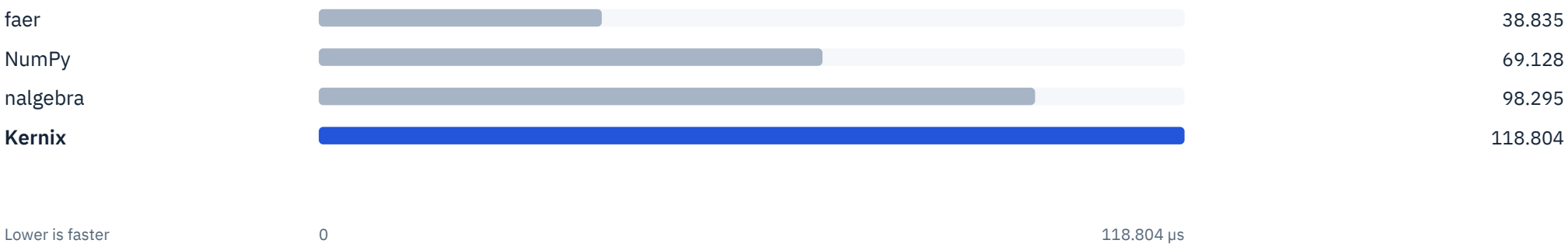


Kernix 0.10.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



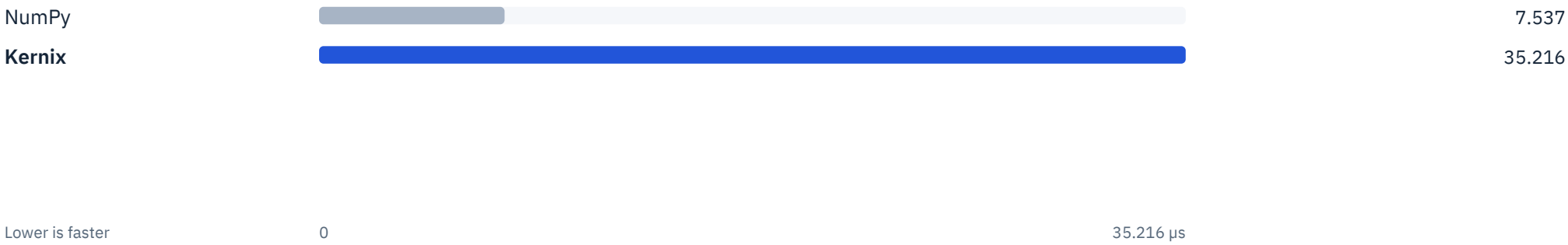
Kernix 0.10.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.10.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



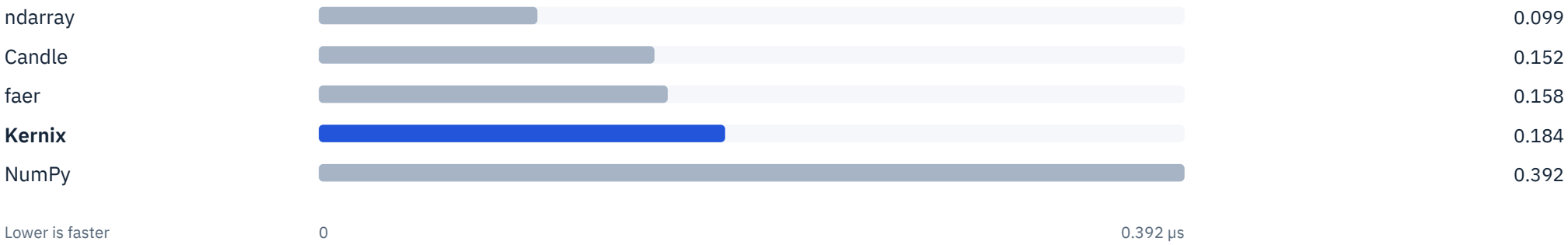
Kernix 0.10.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	74000	74
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	113635	113.635
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	39301	39.301
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	34236	34.236
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1676	1.676
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2675	2.675
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1509	1.509
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1504	1.504
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1489	1.489
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1520	1.52

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11013	11.013
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10025	10.025
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9579	9.579
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	10680	10.68
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11344	11.344
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10864	10.864
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	82695	82.695
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	78010	78.01
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	112473	112.473
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	80246	80.246

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	126260.99999999999	126.26099999999999
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	76016	76.016
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	564860	564.86
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	599005	599.005
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	665835	665.835
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	654160	654.16
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	566777	566.777
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	725077	725.077
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6062	6.062
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4719	4.719

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5174	5.174
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6270	6.27
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4577	4.577
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	5806	5.806
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	39879	39.879
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51568	51.568
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	38177	38.177
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	37724	37.724
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	48494	48.494
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	42098	42.098

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	309202	309.202
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	294419	294.419
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	407021	407.021
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	280464	280.464
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	312027	312.027
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	392602	392.602
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1936.0000000000002	1.9360000000000002
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3054	3.054
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1886	1.886
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	9672	9.672

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	41284	41.284
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	294668	294.668
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	39180	39.18
Cholesky factorization / nalgebra 512 × 512 · f64	6158120	6158.12
Cholesky factorization / NumPy 512 × 512 · f64	4039864	4039.864
Cholesky factorization / faer 512 × 512 · f64	1541686	1541.686
Cholesky factorization / Kernix 512 × 512 · f64	5157487	5157.487
Column-pivoted QR / Kernix 96 × 32 · f64	53411	53.411
Column-pivoted QR / faer 96 × 32 · f64	71009	71.009
Column-pivoted QR / nalgebra 96 × 32 · f64	73743	73.743

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	446662	446.662
Column-pivoted QR / nalgebra 256 × 64 · f64	781249	781.249
Column-pivoted QR / Kernix 256 × 64 · f64	519996	519.996
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	73676	73.676
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	64624	64.624
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	43150	43.15
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	141059	141.059
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	119212	119.212
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	56992	56.992
Thin QR / Kernix 96 × 32 · f64	44501	44.501

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	65412	65.412
Thin QR / NumPy 96 × 32 · f64	40505	40.505
Thin QR / nalgebra 96 × 32 · f64	37932	37.932
Thin QR / faer 256 × 64 · f64	406431	406.431
Thin QR / Kernix 256 × 64 · f64	485343	485.343
Thin QR / NumPy 256 × 64 · f64	377160	377.16
Thin QR / nalgebra 256 × 64 · f64	374058	374.058
Cholesky factorization / NumPy 64 × 64 · f64	10821	10.821
Cholesky factorization / nalgebra 64 × 64 · f64	13622	13.622
Cholesky factorization / faer 64 × 64 · f64	8319	8.319

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	19580	19.58
Cholesky factorization / Kernix 128 × 128 · f64	118804	118.804
Cholesky factorization / nalgebra 128 × 128 · f64	98295	98.295
Cholesky factorization / faer 128 × 128 · f64	38835	38.835
Cholesky factorization / NumPy 128 × 128 · f64	69128	69.128
Cholesky factorization / faer 256 × 256 · f64	194443	194.443
Cholesky factorization / Kernix 256 × 256 · f64	755046	755.046
Cholesky factorization / nalgebra 256 × 256 · f64	691424	691.424
Cholesky factorization / NumPy 256 × 256 · f64	522822	522.822
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3848.0000000000005	3.8480000000000005

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6053	6.053
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2589	2.589
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	5091	5.091
Model inference / Candle batched normalized/sliced MLP · f64	3370	3.37
Model inference / Kernix batched normalized/sliced MLP · f64	2155	2.155
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2794	2.794
Model recording + forward / Candle batched normalized/sliced MLP · f64	5180	5.18
Model backward / Kernix prepared graph → fresh gradients · f64	3151	3.151
Model backward / Candle prepared graph → fresh gradients · f64	14237	14.237
Parameter update / Kernix transactional SGD, fixed gradients · f64	526	0.526

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	35216	35.216
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	7537	7.537
Row softmax / Candle 256 × 64 · f64	88328	88.328
Row softmax / ndarray 256 × 64 · f64	72742	72.742
Row softmax / Kernix 256 × 64 · f64	85131	85.131
Row softmax / NumPy 256 × 64 · f64	99236	99.236
Create zeros / Kernix 64 × 64 · f32	184	0.184
Create zeros / faer 64 × 64 · f32	158	0.158
Create zeros / ndarray 64 × 64 · f32	99	0.099
Create zeros / Candle 64 × 64 · f32	152	0.152

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	392	0.392
Create ones / Candle 64 × 64 · f32	216	0.216
Create ones / NumPy 64 × 64 · f32	1281	1.281
Create ones / Kernix 64 × 64 · f32	226	0.226
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / faer 64 × 64 · f32	201	0.201